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Inventor(s)

Byrne; Donny M.

METHOD AND SYSTEM FOR PUTTING STROKE IMPROVEMENT

Abstract

A system for putting stroke improvement includes a plurality of arches, a plurality of rods extending through holes of the plurality of arches to provide structural support to the plurality of arches so that the plurality of arches are stood upright, and a guide wire that extends from a first arch of the plurality of arches to a second arch of the plurality of arches. The plurality of rods includes two rods that pass through an upper portion of the plurality of arches and that are positioned on either side of the guide wire when the guide wire is viewed from above the system.

Inventors: Byrne; Donny M. (Montgomery, TX)

Applicant: Byrne; Donny M. (Montgomery, TX)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application claims priority from, and incorporates by reference the entire disclosure of, U.S. Provisional Patent Application No. 63/558,750 filed on Feb. 28, 2024.

TECHNICAL FIELD

[0002] The present disclosure relates generally to golf and more particularly, but not by way of limitation, to a method and system for improving a player's putting stroke.

BACKGROUND

[0003] This section provides background information to facilitate a better understanding of the various aspects of the disclosure. It should be understood that the statements in this section of this document are to be read in this light, and not as admissions of prior art.

[0004] Golf is a sport embraced by many people, from weekend amateurs to professionals. There are many aspects to the game of golf that require great skill to master. Putting is a critical part of the game of golf and requires great precision and focus. To master putting, a player must have precise, repeatable movements. Precision requires that the face of the putter that strikes the ball be very carefully aligned in multiple planes. Without the aid of a training device, it can be very difficult for a player to see and understand the alignment of the putter face upon striking the ball.

SUMMARY OF THE INVENTION

[0005] This summary is provided to introduce a selection of concepts that are further described below in the Detailed Description. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it to be used as an aid in limiting the scope of the claimed subject matter.

[0006] The system described herein provides a way for a player to see various aspects of putting in a way that provides instant feedback to the quality of their putting stroke. These aspects include the alignment of the putter face relative to the ball, and the alignment of the stroke throughout the backswing and follow through. Described herein is a system that provides visual feedback to a user of the alignment of the putter throughout the user's putting stroke.

[0007] In various aspects, a system for putting stroke improvement includes a plurality of arches, a plurality of rods extending through holes of the plurality of arches to provide structural support to the plurality of arches so that the plurality of arches are stood upright, and a guide wire that extends from a first arch of the plurality of arches to a second arch of the plurality of arches. The plurality of rods includes two rods that pass through an upper portion of the plurality of arches and that are positioned on either side of the guide wire when the guide wire is viewed from above the system.

[0008] In some aspects, the system includes an alignment plate that is slidably mounted to the two rods between two arches of the plurality of arches. In some aspects, the alignment plate includes an alignment mark that is configured to align with the guide wire.

[0009] In some aspects, the system includes a pair of stroke length markers slidably mounted to one rod of the two rods, wherein the pair of stroke length markers are positioned on opposite sides of the alignment plate.

[0010] In some aspects, each arch of the plurality of arches comprises a different-sized opening through which a golf ball may pass.

[0011] In some aspects, each arch of the plurality of arches comprises a same-sized opening through which a golf ball may pass.

[0012] In some aspects, the guide wire is secured to two arches of the plurality of arches by golf tees that are inserted through loops at the ends of the guide wire.

[0013] In some aspects, each rod of the plurality of rods is an alignment stick.

[0014] In some aspects, the system includes a spy glass configured to be placed on top of a pair of

rods of the plurality of rods, the spy glass comprising a pair of orthogonally arranged alignment lines that form a crosshair. In some aspects, one line of the pair of orthogonally arranged alignment lines is oriented parallel to the plurality of rods. In some aspects, the spy glass comprises a pair of grooves that are configured to receive a pair of rods of the plurality of rods. In some aspects, when the guide wire and one alignment line of the pair of orthogonally arranged alignment lines are vertically aligned, an eye of a user standing above the guide wire and the one alignment line is positioned directly over a ball that is positioned directly below the crosshair.

[0015] In some aspects, the system includes a spy glass configured to be placed on top of a pair of rods of the plurality of rods, the spy glass comprising a pair of grooves that are configured to receive a pair of rods of the plurality of rods.

[0016] In some aspects, the holes of the plurality of arches through which rods pass comprise o-rings to more securely hold the rods.

[0017] In various aspects, a stimp meter includes a pair of rods and an arch comprising a plurality of holes, each rod of the pair of rods extending through one hole of the plurality of holes. The rods are arranged to be parallel to form a ramp down which a ball can roll.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] A more complete understanding of the subject matter of the present disclosure may be obtained by reference to the following Detailed Description when taken in conjunction with the accompanying Drawings wherein:

[0019] FIGS. 1 and 2 are perspective views of a system for improving a player's putting stroke, according to aspects of the disclosure;

[0020] FIG. 3 is a top view of the system of FIGS. 1 and 2, according to aspects of the disclosure;

[0021] FIG. 4 is a close-up view of an alignment plate of the system of FIGS. 1 and 2, according to aspects of the disclosure;

[0022] FIGS. 5A and 5B are close-up views of arches of the system of FIGS. 1 and 2, according to aspects of the disclosure;

[0023] FIG. 6 is a close-up view of ends of a guide wire of the system of FIGS. 1 and 2, according to aspects of the disclosure;

[0024] FIG. 7 illustrates arches of the system of FIGS. 1 and 2, according to aspects of the disclosure;

[0025] FIG. 8 illustrates a free-standing arch, according to aspects of the disclosure;

[0026] FIG. 9 is a perspective view of a stimp meter configuration, according to aspects of the disclosure;

[0027] FIGS. 10A and 10B are top-down views of a spy glass, according to aspects of the disclosure; and

[0028] FIG. 11 is a close-up view of an arch, according to aspects of the disclosure.

DETAILED DESCRIPTION

[0029] It is to be understood that the following disclosure provides many different embodiments, or examples, for implementing different features of various embodiments. Specific examples of components and arrangements are described below to simplify the disclosure. These are, of course, merely examples and are not intended to be limiting. The section headings used herein are for organizational purposes and are not to be construed as limiting the subject matter described.

Reference will now be made to more specific embodiments of the present disclosure and data that provides support for such embodiments. However, it should be noted that the disclosure below is for illustrative purposes only and is not intended to limit the scope of the claimed subject matter in any way.

[0030] FIGS. 1 and 2 are perspective views of a system **100** for improving a player's putting stroke, according to aspects of the disclosure. FIG. 3 is a top view of the system of FIGS. 1 and 2. System **100** includes a plurality of supports or rods **102** that extend between and through a plurality of arches **104(1)**, **104(2)**, and **104(3)**. FIGS. 1-3 illustrate four rods **102**, but it will be appreciated that fewer than four rods **102** may be used. For example, system **100** as shown includes two rods **102** that are positioned close to the ground and two rods **102** that are positioned close to the tops of arches **104(1)**, **104(2)**, and **104(3)**. In some aspects, the lower rods **102** close to the ground may be omitted, leaving the two upper rods **102** positioned close to the tops of arches **104(1)**, **104(2)**, and **104(3)**. In some aspects, rods **102** may be alignment sticks that are commonly owned and used by golfers. In other aspects, rods **102** can be any suitable rod member, such as a dowel rod and the like. In a typical aspect, upper rods **102** are spaced to be relatively close together (e.g., such that a width of rods **102** is less than the width of a golf ball so that a putter's shaft is not obstructed by rods **102** while putting).

[0031] FIGS. 1-3 illustrate three arches **104**, but it will be appreciated that less than three or more than three arches **104** may be used as desired. Each arch **104** includes an opening **119** through which a golf ball may pass. Arches **104(1)**, **104(2)**, and **104(3)** each have different sized openings **119(1)**, **119(2)**, **119(3)** (best seen in FIG. 7). The different sized openings allow the user to select different arches to change the difficulty or precision that is needed to successfully pass the golf ball through the arch. The order of the arches along a length of system **100** may be chosen as desired. Each arch **104** includes a plurality of holes **105** through which the plurality of rods **102** may pass (see FIG. 7). As shown in FIGS. 1-3, each arch **104** includes four holes **105** for rods **102**. Each arch **104** additionally includes a hole **107** through which a guide wire **106** may pass. It will be appreciated that various aspects of system **100** can be changed or modified without departing from the scope of the invention. For example, system **100** can include more or fewer arches **104** (e.g., as few as two and more than three). The arches used can have different sized openings (as shown in the Figures discussed hereinabove), or same sized openings.

[0032] System **100** includes a pair of stroke length markers **108(1)** and **108(2)**. Markers **108(1)** and **108(2)** are slidably arranged on one rod **102** of the plurality of rods **102** and provide a visual reference points for the end of a back swing (**108(1)**) and the end of a follow through (**108(2)**). A user can move markers **108(1)** and **108(2)** closer together for a softer stroke and farther apart for a more powerful stroke.

[0033] System **100** includes an alignment plate **110** that provides a visual indication to a user that the face of their putter is squarely aligned with the golf ball. Alignment plate **110** includes two holes **111** through which the two rods **102** that are positioned close to the tops of arches **104** pass through. Alignment plate **110** is positioned between markers **108(1)** and **108(2)** and is configured so that it can slide along rods **102** to be positioned as desired by the user. Alignment plate **110** also includes an alignment mark **113** that is positioned on a top side thereof. Alignment mark **113** may be a notch, a colored mark, or a combination of the two and is positioned at a midpoint of alignment plate **110** so as to be in the middle of the two rods **102** when a user looks down with their eyes positioned directly over the golf ball (see FIG. 3). In use, the user aligns the face of the putter with the face of alignment plate **110** and the middle of the face of the putter with alignment mark **113** and with guide wire **106**. The use of both guide wire **106** and alignment plate **110** is particularly beneficial as it eliminates alignment errors due to parallax. Using guide wire **106** in between the two rods **102** is beneficial because it allows the user to see the putter more clearly due to it's relatively thin diameter and as the two rods **102** are not blocking the user's view of the center of the putter.

[0034] FIG. 4 is a close-up view of guide plate of the system of FIGS. 1 and 2, according to aspects of the disclosure. FIG. 4 illustrates that guide wire **106** is spaced above the top of alignment plate **110**. This spacing helps improve the alignment of the face of the putter with ball by eliminating errors due to parallax. FIG. 4 illustrates that alignment mark **113** includes both a notch and a dark

marking in the notch.

[0035] FIGS. 5A and 5B are close-up views of arches **104(1)** and **104(2)** of system **100**, according to aspects of the disclosure. FIGS. 5A and 5B illustrate how guide wire **106** is securely affixed to arches **104(1)** and **104(3)** to maintain a bit of tension in guide wire **106**. Each arch **104** includes hole **107**. To affix guide wire **106** to an arch **102**, the user threads an end of guide wire **106** through hole **107**. Guide wire **106** includes a loop **115(1)** and a loop **115(2)** (see FIG. 6) at its ends that are sized to receive a shaft of a golf tee **117**. Once an end of guide wire **106** has been threaded through a hole **107**, golf tee **117** is inserted into hole **107**. This is repeated for the opposite end of guide wire **106**. With golf tees **117** inserted into holes **107**, arches **104(1)** and **104(3)** may be gently slid apart to tension guide wire **106**. In other aspects, arches **104(1)** and **104(3)** may include a feature, such as a hook or tab, to which loops **115(1)** and **115(2)** may be attached to secure guide wire **106**.

[0036] FIG. 7 illustrates arches **104(1)**, **104(2)**, and **104(3)** of system **100**, according to aspects of the disclosure. As shown in FIG. 7, each arch **104(1)**, **104(2)**, **104(3)** has a different-sized opening **119(1)**, **119(2)**, **119(3)**, respectively. The different-sized openings allow the user to place arches **104(1)**, **104(2)**, and **104(3)** as desired to putt through an easier (larger) opening (**119(3)**) or a more difficult (smaller) opening (**119(1)**).

[0037] A method of using system **100** will now be discussed with reference to FIGS. 1-7. To practice their putting stroke, the user places a golf ball between two arches **104** and directly beneath guide wire **106** as shown in FIG. 3. Alignment plate **110** is then positioned to be aligned with a backside edge of the golf ball (i.e., the portion of the golf ball that the face of the putter will contact). Stroke length markers **108(1)** and **108(2)** are then positioned as desired on either side of alignment plate **110** for the stroke length that the user is practicing. For example, for a shorter putt, a shorter stroke length is appropriate and markers **108(1)** and **108(2)** are placed relatively closer together than for a longer stroke length.

[0038] To begin the putt, the user places the putter behind the golf ball and aligns the face of the putter to be coplanar with the face of alignment plate **110**, while at the same time making sure that the center of the putter face is in-line with guide wire **106** and alignment mark **113** of alignment plate **110**. Aligning the face of the putter with the face of alignment plate **110** squares the putter face so that the golf ball is not pushed or pulled relative to the intended trajectory of the golf ball. The use of these two different vertically aligned reference points eliminates errors caused by parallax and reinforces to the user that the putter face is properly aligned for the putt so that the golf ball will be struck squarely. After the user has properly aligned the putter, the user draws the putter back to the marker **108(1)** and then follows through the golf ball to marker **108(2)**. During the backstroke and follow through, the user's goal is to watch the center of the putter face relative to guide wire **106** and to attempt to keep the two aligned throughout the entire stroke. If alignment is maintained for the entire stroke, the user is assured that the golf ball will be struck squarely. In some aspects, it is helpful for the putter to include an alignment line so that the user can visibly see and track the alignment of the putter relative to guide wire **106** (i.e., the alignment line on the putter remains parallel with guide wire **106** throughout the stroke).

[0039] FIG. 8 illustrates a free-standing arch **112**, according aspects of the disclosure. Free-standing arch **112** includes one arch **104** of the arches **104(1)**-**104(3)** and a pair of feet **114**. Each foot **114** includes a slot **116** that is configured to receive a base of free-standing arch **112**. Each foot **114** also includes one or more holes **118** that are sized to receive a golf tee so that foot **114** may be staked to the ground to secure free-standing arch **112** to the ground. Free-standing arch **112** may be used for putting practice as a sort of make-shift hole or target. Free-standing arch **112** may be constructed using any one of arches **104(1)**-**104(3)** discussed above, giving the user the ability to choose which size opening **119** to use for practice.

[0040] FIG. 9 is a perspective view of a stimpmeter configuration **150**, according to aspects of the disclosure. Stimpmeter configuration **150** includes two rods **102** and one arch **104** of the arches **104(1)**-**104(3)**. The two rods **102** are inserted through the two upper holes **105** as shown in FIG. 9.

Stimpmeter configuration **150** can be used with a golf ball to get the read of a putting surface (e.g., a green). The user places the golf ball on the ramp formed by the two rods **102** and allows it to roll down the ramp and onto the putting surface. The user then watches the path of the golf ball to understand how the ball will roll across the putting surface in response to undulations, imperfections, etc. of the putting surface. Stimpmeter configuration **150** is a useful tool because it provides a “true” trajectory of the golf ball that is not influenced or compromised by an inaccurate striking of the golf ball by the putter face. This allows the user to get a more accurate view of how the golf ball will travel across the putting surface if the user strikes the golf ball with the putter using the proper alignment.

[0041] FIGS. **10A** and **10B** are top-down views of spy glasses for use with, for example, system **100**, according to aspects of the disclosure. FIG. **10A** illustrates a spy glass **160** that is shown resting on top of the upper rods **102**. In a typical aspect, spy glass **160** is made from a transparent material so that the user can see a ball positioned under spy glass **160** when the user's eyes are positioned directly above the ball. Spy glass **160** includes a pair of grooves **162** that are formed into a bottom face of spy glass **160**. The pair of grooves **162** are configured to complement the shape of rods **102** so that spy glass **160** aligns or indexes with the pair of rods **102** (e.g., the width or radius of a groove **162** complements the width or radius of a rod **102**, and the distance between the pair of grooves **162** matches the distance between the rods **102** so that spy glass **162** positively sits in place upon the rods **102**).

[0042] Spy glass **160** further includes an alignment line **164** and an alignment line **166**. Alignment line **164** is arranged parallel to grooves **162** and rods **102**, and is positioned to be equidistant from grooves **162**. Alignment line **166** is arranged perpendicular to alignment line **164** to form a crosshair therewith. The crosshair is used in conjunction with guide wire **106** to provide a visual indication to a user that the user's eyes are directly over a golf ball positioned thereunder. For example, as shown in FIG. **10A**, guide wire **106** is arranged parallel to rods **102** and slightly above rods **102** (e.g., 0.5 inches to 2 inches above, though the vertical distance may differ if desired). Spy glass **160** rests on top of rods **102**, and beneath guide wire **106**. The vertical spacing between spy glass **160** and guide wire **106** allows a user to visually see when the user's eyes are directly over a ball positioned beneath the crosshair by aligning guide wire **106** with alignment line **164** (as shown in FIG. **10A**). Without the vertical alignment aspect of guide wire **106** relative to spy glass **160**, it is more difficult for the user to be certain that the user's eyes are positioned directly over the ball.

[0043] In addition to forming the crosshair, alignment line **166** provides the user with a reference for the face of the putter as the putter strikes the ball (e.g., the user attempts to align the face of the putter with alignment line **166** as the putter strikes the ball so that the putter face is square to the ball as contact is made with the ball).

[0044] Spy glass **160** is also configured with edges **167** to be relatively close to rods **102**. The configuration of edges **167** in this manner allows maximum clearance for a shaft of the putter relative to rods **102**. This can be particularly important for certain putter designs in which the putter shaft mates to the head of the putter in the middle of the head. FIG. **10B** illustrates an alternate configuration of a spy glass **161** in which spy glass **161** has a rounded shape without edges **167**. Apart from the lack of edges **167**, spy glass **161** is otherwise similar to spy glass **160** and the discussion of spy glass **160** similarly applies to spy glass **161**. It will be appreciated that spy glasses **160**, **161** could be configured to have other shapes. For example, spy glass **161** is shown in FIG. **10B** as being generally circular. In other aspects, spy glass **161** could comprise other shapes, including and not limited to, square, rectangle, oval, geometric, and the like. The shape of the spy glass may be configured as desired so long as grooves **162** and alignment lines **164**, **166** are incorporated as discussed above.

[0045] FIG. **11** is a close-up view of arch **104** in an alternate configuration, according to aspects of the disclosure. Arch **104** of FIG. **11** has been configured to include grommets or o-rings **168** in holes **107(2)** and **107(3)**. O-rings **168** more securely receive rods **102**, improving the stability of

system **100** (e.g., arches **104** are less prone to sliding relative to rods **104**). In some aspects, o-rings **168** may be sized to have an internal diameter that is less than an external diameter of rods **102** to form a force fit therebetween. In other aspects, the internal diameter of o-rings **168** may be sized to be equal to or greater than the external diameter of rods **102**. In the aspect of FIG. **11**, hole **107(1)** has also been reduced in diameter to more closely complement the diameter of guide wire **106**. [0046] Although various embodiments of the present disclosure have been illustrated in the accompanying Drawings and described in the foregoing Detailed Description, it will be understood that the present disclosure is not limited to the embodiments disclosed herein, but is capable of numerous rearrangements, modifications, and substitutions without departing from the spirit of the disclosure as set forth herein.

[0047] The term “substantially” is defined as largely but not necessarily wholly what is specified, as understood by a person of ordinary skill in the art. In any disclosed embodiment, the terms “substantially”, “approximately”, “generally”, and “about” may be substituted with “within [a percentage] of” what is specified, where the percentage includes 0.1, 1, 5, and 10 percent.

[0048] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the disclosure. Those skilled in the art should appreciate that they may readily use the disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the disclosure. The scope of the invention should be determined only by the language of the claims that follow. The term “comprising” within the claims is intended to mean “including at least” such that the recited listing of elements in a claim are an open group. The terms “a”, “an”, and other singular terms are intended to include the plural forms thereof unless specifically excluded.

[0049] Conditional language used herein, such as, among others, “can”, “might”, “may”, “e.g.”, and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0050] While the above detailed description has shown, described, and pointed out novel features as applied to various embodiments, it will be understood that various omissions, substitutions, and changes in the form and details of the devices or algorithms illustrated can be made without departing from the spirit of the disclosure. As will be recognized, the processes described herein can be embodied within a form that does not provide all of the features and benefits set forth herein, as some features can be used or practiced separately from others. The scope of protection is defined by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

[0051] Although various embodiments of the method and apparatus of the present invention have been illustrated in the accompanying Drawings and described in the foregoing Detailed Description, it will be understood that the invention is not limited to the embodiments disclosed, but is capable of numerous rearrangements, modifications and substitutions without departing from the spirit of the invention as set forth herein.

Claims

- 1.** A system for putting stroke improvement comprising: a plurality of arches; a plurality of rods extending through holes of the plurality of arches to provide structural support to the plurality of arches so that the plurality of arches are stood upright; and a guide wire that extends from a first arch of the plurality of arches to a second arch of the plurality of arches, wherein the plurality of rods includes two rods that pass through an upper portion of the plurality of arches and that are positioned on either side of the guide wire when the guide wire is viewed from above the system.
 - 2.** The system of claim 1, further comprising an alignment plate that is slidably mounted to the two rods between two arches of the plurality of arches.
 - 3.** The system of claim 2, wherein the alignment plate includes an alignment mark that is configured to align with the guide wire.
 - 4.** The system of claim 1, further comprising a pair of stroke length markers slidably mounted to one rod of the two rods, wherein the pair of stroke length markers are positioned on opposite sides of the alignment plate.
 - 5.** The system of claim 1, wherein each arch of the plurality of arches comprises a different-sized opening through which a golf ball may pass.
 - 6.** The system of claim 1, wherein each arch of the plurality of arches comprises a same-sized opening through which a golf ball may pass.
 - 7.** The system of claim 1, wherein the guide wire is secured to two arches of the plurality of arches by golf tees that are inserted through loops at the ends of the guide wire.
 - 8.** The system of claim 1, wherein each rod of the plurality of rods is an alignment stick.
 - 9.** The system of claim 1, further comprising a spy glass configured to be placed on top of a pair of rods of the plurality of rods, the spy glass comprising a pair of orthogonally arranged alignment lines that form a crosshair.
 - 10.** The system of claim 9, wherein one line of the pair of orthogonally arranged alignment lines is oriented parallel to the plurality of rods.
 - 11.** The system of claim 9, wherein the spy glass comprises a pair of grooves that are configured to receive a pair of rods of the plurality of rods.
 - 12.** The system of claim 9, wherein when the guide wire and one alignment line of the pair of orthogonally arranged alignment lines are vertically aligned, an eye of a user standing above the guide wire and the one alignment line is positioned directly over a ball that is positioned directly below the crosshair.
 - 13.** The system of claim 1, further comprising a spy glass configured to be placed on top of a pair of rods of the plurality of rods, the spy glass comprising a pair of grooves that are configured to receive a pair of rods of the plurality of rods.
 - 14.** The system of claim 1, wherein the holes of the plurality of arches through which rods pass comprise o-rings to more securely hold the rods.
 - 15.** A stimpmeter comprising: a pair of rods; and an arch comprising a plurality of holes, each rod of the pair of rods extending through one hole of the plurality of holes, wherein the rods are arranged to be parallel to form a ramp down which a ball can roll.
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